MANUFACTURING COST AND PROFITABILITY ANALYSIS

1. Problem Statement

Develop a manufacturing cost analysis system that calculates material costs, wastage costs, labour charges, and machine operating costs for a production batch. The system determines the total production cost, cost per unit, selling revenue, and profit. It then checks whether the profit meets the specified target.

2. Objective

To calculate the complete cost of manufacturing, determine the selling price and expected profit, and identify whether the production batch achieves the required profit target.

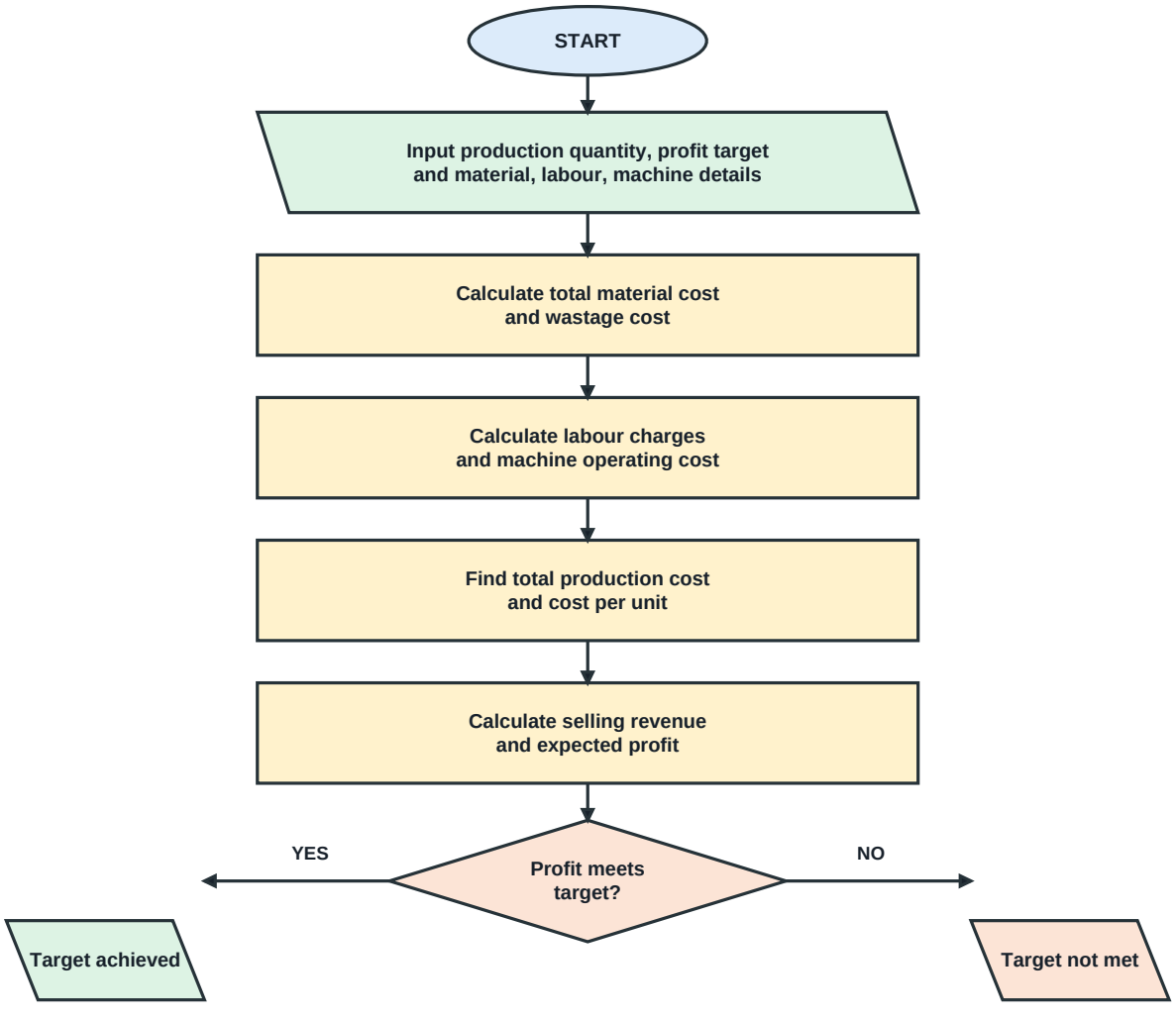
3. Assumptions

1. All costs and quantities are entered by the user.
2. Wastage cost is calculated as a percentage of material cost.
3. Labour cost is calculated using labour hours and hourly wage.
4. Machine cost is calculated using machine hours and hourly operating rate.
5. Selling price is entered per finished unit, and profit target is a total batch amount.

4. Algorithm

1. Start.
2. Read production quantity, material quantity and rate, wastage percentage, labour hours and wage, machine hours and rate, selling price per unit, and profit target.
3. Calculate material cost.
4. Calculate wastage cost.
5. Calculate labour and machine costs.
6. Add all costs to find total production cost.
7. Calculate cost per unit and total selling revenue.
8. Calculate profit by subtracting total production cost from selling revenue.
9. Compare profit with the target and display the result.
10. Stop.

5. Flowchart (Clear Normal Symbols)



6. Pseudocode

START
INPUT production_quantity
INPUT material_quantity, material_rate, wastage_percent
INPUT labour_hours, labour_wage
INPUT machine_hours, machine_rate
INPUT selling_price_per_unit, profit_target
material_cost = material_quantity * material_rate
wastage_cost = material_cost * wastage_percent / 100
labour_cost = labour_hours * labour_wage
machine_cost = machine_hours * machine_rate
total_cost = material_cost + wastage_cost + labour_cost + machine_cost
cost_per_unit = total_cost / production_quantity
revenue = production_quantity * selling_price_per_unit
profit = revenue - total_cost
IF profit >= profit_target THEN
  DISPLAY "Profit target achieved"
ELSE
  DISPLAY "Profit target not achieved"
END IF
DISPLAY all cost details, revenue and profit
STOP

7. Python Code

```
print("MANUFACTURING COST AND PROFITABILITY ANALYSIS")

production_quantity = int(input("Enter production quantity: "))
material_quantity = float(input("Enter material quantity used: "))
material_rate = float(input("Enter material cost per unit: "))
wastage_percent = float(input("Enter wastage percentage: "))

labour_hours = float(input("Enter total labour hours: "))
labour_wage = float(input("Enter labour cost per hour: "))

machine_hours = float(input("Enter machine operating hours: "))
machine_rate = float(input("Enter machine cost per hour: "))

selling_price = float(input("Enter selling price per product: "))
profit_target = float(input("Enter required profit target: "))

material_cost = material_quantity * material_rate
wastage_cost = material_cost * wastage_percent / 100
labour_cost = labour_hours * labour_wage
machine_cost = machine_hours * machine_rate

total_cost = (material_cost + wastage_cost +
              labour_cost + machine_cost)

cost_per_unit = total_cost / production_quantity
revenue = production_quantity * selling_price
profit = revenue - total_cost

print("\\n--------- COST ANALYSIS ---------")
print("Material cost: Rs.", round(material_cost, 2))
print("Wastage cost: Rs.", round(wastage_cost, 2))
print("Labour cost: Rs.", round(labour_cost, 2))
print("Machine cost: Rs.", round(machine_cost, 2))
print("Total production cost: Rs.", round(total_cost, 2))
print("Cost per unit: Rs.", round(cost_per_unit, 2))
print("Total selling revenue: Rs.", round(revenue, 2))
print("Profit: Rs.", round(profit, 2))

if profit >= profit_target:
    print("Profit target achieved")
else:
    print("Profit target not achieved")
```

8. Expected Output

MANUFACTURING COST AND PROFITABILITY ANALYSIS
Enter production quantity: 100
Enter material quantity used: 120
Enter material cost per unit: 20
Enter wastage percentage: 5
Enter total labour hours: 40
Enter labour cost per hour: 100
Enter machine operating hours: 20
Enter machine cost per hour: 150
Enter selling price per product: 500
Enter required profit target: 10000

--------- COST ANALYSIS ---------

Material cost: Rs. 2400.0
Wastage cost: Rs. 120.0
Labour cost: Rs. 4000.0
Machine cost: Rs. 3000.0
Total production cost: Rs. 9520.0
Cost per unit: Rs. 95.2
Total selling revenue: Rs. 50000.0
Profit: Rs. 40480.0
Profit target achieved